

Ashu Pabreja

Aspiring Data Scientist — Generative AI Engineer & Researcher — Business Development

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Profile Summary

- AI & Data Science professional with over one year of hands-on experience and a strong academic foundation in Computer Science and Engineering, specializing in Machine Learning, Generative AI, NLP, Cloud Computing and data-driven system design across FinTech, AgriTech, GovTech.
- Strong ability to translate business and operational problems into scalable AI and analytics solutions.
- Hands-on experience in Business Development, including lead generation strategies, CRM systems, solution architecture planning, and sales workflow optimization using lead generation platforms.
- Recognized for public speaking, adaptability, and problem-solving skills, with the ability to effectively bridge technical solutions and business needs.

Education

• Msc in Data Science and AI	Middlesex University Dubai	Jan 2026 - Dec 2027
• B.Tech in Computer Science and Engineering PPI: 7.87	Institute of Technology, Nirma University	October 2021 - May 2025
• Class 12 (Science) Percentage: 91.23%	C.D Patel High School, Talod	July 2019 - July 2021

Areas of Expertise

- Artificial Intelligence & Machine Learning
- Cloud Computing & CI/CD Pipelines
- Langchain, MCP, AI agents
- Vector Databases (FAISS, Pinecone), Multimodal Embeddings, Prompt Engineering
- Business Development
- Research, Experimentation & Technical Documentation
- Model Training, Evaluation & Optimization
- Database Management (SQL & NoSQL)
- Cross-functional Collaboration
- Generative AI & NLP
- Team Building & Leadership
- Software Development & System Design
- Continuous Learning & Knowledge Sharing

Skills

Technical Skills

- Python, JavaScript
- Machine Learning, Deep Learning, NLP, Generative AI
- Data Analytics & Visualization
- Git, Docker, Hugging Face, VS Code, AWS, Vertex AI (Foundational)
- Microservice Architecture and Programming

Soft Skills

- Adaptability & Resilience
- Creative & Innovative Thinking
- Leadership & Social Influence
- Teamwork & Effective Communication

Project Experience (Academic & freelance)

Plant Disease Classification

AgriTech / Precision Agriculture

- Built a ResNet-based deep learning model using Python, TensorFlow, and Keras for multi-class plant disease classification.
- Trained and evaluated multiple ResNet family architectures, achieving **95% accuracy**.
- Implemented data preprocessing, visualization, and model evaluation using NumPy, Pandas, and Matplotlib.
- Enabled early disease detection to reduce crop loss and improve agricultural decision-making.

MoneyManagers – Financial Analytics Dashboard

FinTech / Corporate Finance

- Developed a full-stack MERN dashboard for real-time tracking of revenue, expenses, and sales performance.
- Integrated machine learning-based forecasting to predict future revenue trends and financial outcomes.
- Built interactive visualizations to improve financial visibility and executive-level decision-making.
- Reduced manual reporting effort by **60%** through automated analytics and predictive insights.

Career Counselling Chatbot (Gujarat (Indian) Government Project)

EdTech / GovTech

- Designed a multilingual AI-powered chatbot using LangChain and LLMs to automate student admission and counselling workflows.
- Implemented a RAG pipeline with document chunking, embeddings, and vector stores for accurate, context-aware responses.
- Enabled smart document processing via OCR (PDFs and images) and secure user authentication using JWT.

- Reduced administrative workload while improving accessibility, response consistency, and scalability.
- LinkedIn AI Career Assistant** *HRTech / Talent Analytics*
- Built an AI-driven assistant to analyze LinkedIn profiles and deliver personalized career insights using LLMs.
 - Implemented job-fit analysis, skill-gap detection, and career path forecasting aligned with industry trends.
 - Enabled conversational career guidance through natural language interaction and real-time recommendations.
 - Improved profile optimization and candidate-to-role alignment through data-driven insights.
- Industrial IoT for Real-Time Monitoring (OPMS)** *Manufacturing / Industrial Automation*
- Designed a secure cloud-based Industrial IoT system integrating SCADA, DCS, and plant sensors for real-time monitoring.
 - Implemented predictive maintenance using Random Forest and Prophet models to forecast equipment behavior.
 - Built role-based dashboards and automated alerting for anomaly detection and operational insights.
 - Reduced downtime and improved production efficiency through predictive analytics and centralized monitoring.

Previous Work Experience

- Software Engineer** *E-commerce / Digital Solutions*
Digiwagon Technologies Pvt. Ltd., Ahmedabad, India *Sep 2025 – Dec 2025*
- Designed and maintained scalable data pipelines for e-commerce clients, enabling efficient data ingestion, transformation, and analytics-ready storage.
 - Worked on data manipulation, validation, and automation workflows to support reporting, operational insights, and business decision-making.
 - Collaborated with cross-functional teams to ensure data reliability, performance optimization, and smooth integration with frontend systems.

- Business Development Intern** *Industrial Solutions / Manufacturing*
CEREBULB (India) Pvt. Ltd., Gandhinagar, India *Jan 2025 – Jun 2025*
- Designed data-driven solution architectures for industrial clients, aligning business requirements with analytics and automation use cases.
 - Supported lead generation, CRM workflows, and client requirement analysis to improve solution positioning and sales efficiency.
 - Strengthened ability to translate industrial problems into scalable technical solutions, bridging business strategy with data science applications.

- Software Developer Intern** *FinTech / Data Analytics*
Demaze Technologies, Ahmedabad, India *Jun 2024 – Jul 2024*
- Contributed to full-stack development and data-driven features for financial analytics use cases.
 - Worked on the **MoneyManagers Financial Analytics Dashboard**, applying data visualization and predictive analytics concepts to support faster financial reporting.
 - Gained practical exposure to integrating analytics into business-facing applications and improving insight delivery.

- MERN Stack Developer Intern** *Web Applications / SaaS*
Digiwagon Technologies Pvt. Ltd., Ahmedabad, India *Jul 2023 – Aug 2023*
- Developed full-stack web applications using the MERN stack, focusing on frontend optimization, backend APIs, and database performance.
 - Improved understanding of scalable application architecture, RESTful services, and production-ready deployment practices.

Research Work Experience

- Co-authored a peer-reviewed research chapter titled “*Implantable Medical Devices: Navigating Types, Unveiling Risks, and Securing Solutions*” in *Wearable Medical Device Security and Network Protection for Reliable Healthcare* (©2026). Analyzed security risks and protection mechanisms in implantable medical devices. DOI: 10.4018/979-8-3373-4412-6.ch009

Professional Certifications

- AWS Academy Graduate - AWS Academy Cloud Developing
- Agile Project Management Certifications
- AWS Academy Microservices and CI/CD Pipeline Builder
- Explore Generative AI with the Gemini API in Vertex AI
- McKinsey.org Forward Program
- Build Real World AI Applications with Gemini and Imagen – Google Cloud
- Develop GenAI Apps with Gemini and Streamlit – Google Cloud